

The Language Detective: Native Language Identification in English as Second Language Speakers

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Abstract

Cross-linguistic transfer is a key phenomenon in second language acquisition, where the syntactic structures and lexicon of a speaker's first language (L1) influence their use of a second language (L2). This paper explores the application of machine learning (ML) and artificial intelligence (AI) methods in the Native Language Identification (NLI) task, which seeks to automatically classify the native language of an English as a Second Language (ESL) speaker based on their English text. Utilizing the EF-Cambridge Open Language Database (EFCAMDAT), we trained neural network classifiers to accurately identify L1 influences in ESL writings. We developed and tested three models of increasing complexity, each aimed at capturing the subtle patterns that indicate a speaker's native language. Our results demonstrate the feasibility of using ML techniques for this challenging task, contributing to the broader understanding of cross-linguistic influences in second language acquisition. The code for this project is available at <https://github.com/aelashkin/LanguageDetective>.

1 Introduction

Language acquisition is a complex cognitive process that is significantly informed by the speakers native language. This paper addresses the phenomenon known as cross-linguistic transfer, a process particularly evident in second-language learning. The second language (L2) learned and used by non-native speakers is often affected by syntactic structures and lexicon of their first language (L1). This interaction between languages has been an important topic of research in psycholinguistics for many years. More recently, the advances in machine learning (ML) and artificial intelligence (AI) technologies provided a variety of computational methods helpful in addressing cross-linguistic transfer. In Particular, we will leverage modern tools on a Native Language Identification

(NLI) task, a task that seeks to automatically classify the native language of an English as a Second Language (ESL) speaker based on text they have produced in English language.

NLI has significant implications across various domains of language-related research. Successfully performing this task would indicate the existence of distinguishable, L1-influenced patterns in ESL, which could be invaluable for identifying such patterns in future research. However, the task remains challenging due to the complex nature of cross-linguistic influences.

In this project, we aim to advance the state of the art in NLI by training neural network classifiers on a large corpus of data provided by the University of Cambridge (EF Research Lab for Applied Language Learning, University of Cambridge, Faculty of Modern and Medieval Languages and Linguistics, Theoretical and Applied Linguistics Section. Accessed 10.07.2024., n.d.). Our goal is to classify the L1 of the text author with high accuracy while also creating models that are suitable for future interpretability research.

We approach this task with three models of increasing complexity, aiming not only to achieve state-of-the-art performance but also to demonstrate how our results compare to other solutions.

Finally, we provide simple functionality allowing the classification of a single English language text. We do it out of consideration for the practical implications of our work, hoping to provide a tool useful for future research and data collection in both academic and applied contexts.

2 Data

2.1 EFCAMDAT Overview

The EF-Cambridge Open Language Database (EFCAMDAT) is a large-scale corpus of English as a Foreign Language (EFL) learner writings (EF Research Lab for Applied Language Learning, Uni-

versity of Cambridge, Faculty of Modern and Medieval Languages and Linguistics, Theoretical and Applied Linguistics Section. Accessed 10.07.2024., n.d.). EFCAMDAT serves as a valuable resource for research in second language acquisition, language assessment, and computational linguistics and was invaluable for our project.

EFCAMDAT comprises written assignments collected from adult learners enrolled in EF's online English courses worldwide. The dataset captures a wide range of proficiency levels, native language backgrounds, and topical prompts allowing for a diverse set of texts for studying linguistic patterns.

The dataset includes two sheets of examples, main prompts and alternative prompts, collectively amounting to almost 750 thousands rows of data. There are ten native languages included in the data, with unbalanced row counts for each label, favouring Portuguese and Mandarin examples See Table 1

Each text is assigned a CEFR level as an ordered factor, derived from the proficiency levels outlined in the EFCAMDAT guidelines (A1, A2, B1, B2, C1).

Each text is presented in two versions: the original text submitted by the author, denoted as *text*, and a version with spelling corrections applied using the Speller function in Python's autocorrect library, denoted as *text_corrected*. For the purposes of this paper, we primarily focus on the *text_corrected* version. The first two models we propose use the *text_corrected* version due to the simpler tokenization methods employed, which are better suited to the corrected text. We recognize that these models could potentially benefit from using the original texts but chose to prioritize consistency and ease of processing.

The third model, however, utilizes the original text. This model incorporates a more advanced tokenizer that can handle syntactic errors and capitalize on the linguistic signals present in the uncorrected text, thus potentially improving classification outcomes.

2.2 Pre-processing

We put significant effort into the cleaning and preparation of the data for our task. Some broken rows of data, as well as duplicates of the same texts, were identified and removed. However, our central focus was on minimisation of non-linguistic clues. Such clues, while potentially enhancing the performance of the models, would not allow for useful insights into the underlying language patterns.

We identified and replaced revealing geographical references with generic non-identifying words of the same category (for example: "from Germany" would be replaced with "from Country"). Such generalisation would prevent models from learning linguistically uninteresting patterns (people from Germany speak German) while maintaining the ability to contextually embed the text. For examples of such revealing sentences see Appendix A.

3 Related Research

Previous research indicates that native language of an author can indeed be identified from stylistic text features (Koppel et al., 2005). This work emphasized the potential of using a combination of these features to capture L1-specific influences in L2 writing.

As the study of the field progressed, more sophisticated models were attempted. Tetreault et al. (2013) introduced the use of n-grams and part-of-speech (POS) n-grams, which provided a more nuanced representation of the syntactic and lexical patterns characteristic of different L1 groups, highlighting the importance of context and sequence information in capturing cross-linguistic transfer effects.

Furthermore, ensemble methods for the NLI task have been explored and achieved results (Malmasi and Dras, 2018).

In this paper we aim to expand upon the existing research and make steps towards neural network based solution of NLI task.

4 Experiments and Results

Our projects includes three attempted solutions for the problem at hand.

4.1 Baseline Model

To establish a baseline performance threshold we chose to implement a simple, non neural network based, model. We hope it provides some indication of performance improvement when we proceed to further models.

In this baseline scenario we chose to classify the texts in by-sentence manner, where final text prediction is an argmax of summed probabilities for each sentence for each language. We embed each sentence using GloVe embedding method described by (Pennington et al., 2014). Embedded

L1	Examples Count	Nationalities
Arabic	29,292	Saudi Arabian (sa)
French	32,504	French (fr)
German	41,418	German (de)
Italian	35,414	Italian (it)
Japanese	17,084	Japanese (jp)
Mandarin	129,542	Chinese (cn), Taiwanese (tw)
Portuguese	313,508	Brazilian (br)
Russian	49,304	Russian (ru)
Spanish	64,744	Mexican (mx)
Turkish	10,303	Turkish (tr)

Table 1: Counts of examples from each L1 as well as nationalities of origin

sentences are then processed by SGDClassifier implementation from scikit-learn (Pedregosa et al., 2011).

Additionally, we attempted to mitigate the negative impact of class imbalance by adding class weights to the classifier, as well as down-sampling the majority examples.

However, the model failed to demonstrate above-random predictive ability for all combinations of imbalance handling. As seen in Figure 1, model learned to classify most examples as majority label or some other consistent label but didn’t demonstrate a true predictive power.

No significant difference in performance was gleaned from looking at predictions for different language proficiency levels.

4.2 BERT based model

We proceed by attempting to fine-tune a pre-trained BERT model for our task, by adding a feed-forward layer for classification (Devlin, 2018). Specifically, the ‘bert-base-uncased’ model from Hugging Face transformers library (Wolf et al., 2020). For future work, we recommend exploring larger pre-trained models to potentially enhance performance further.

Same set of pre-processing steps was performed as in our baseline model, but this time, we create tokens on text-by-text basis, rather than separating them into individual sentences.

To address the challenge of label imbalance, we experimented with several techniques and found that down-sampling the majority classes was most effective in improving the model’s F1 score, though it led to a slight decrease in accuracy. See Figure 2 for comparison between down-sampled and not down-sampled models. Figure 3 further demonstrates non-trivial generalising ability of our model.

We proceed our evaluation concentrating on the performance of the down-sampled model.

As is shown from Figure 3 left panel, our model of choice performs with above-random accuracy across all L1 labels it has trained upon. Moreover, we argue that most common errors made by our model can be used to strengthen our claim of capturing language based signals in the data. Notably, the model exhibits heightened error rates among linguistically related language pairs, such as Portuguese-Spanish, Portuguese-Italian, French-Italian, and Japanese-Mandarin. Tendency to err within related languages group indicates we have indeed captured underlying linguistic patterns and not some non-linguistic clue we failed to consider.

Further, we evaluate our model’s performance across different language proficiency groups, as our dataset included CERF-level writing proficiency evaluations. Figure 4 provides comparative accuracy scores of test of CERF-based sub-groups. We confirm the significance of association between CERF level and model accuracy using Chi-square test, with p-value close to zero. These test results, combined with empirical performance data, lead us to conclude that the model performs better with lower proficiency levels. This conclusion aligns with our intuition: less proficient ESL speakers tend to exhibit more syntactic and lexical anomalies, making them easier to classify.

For CERF level separated heat-maps see Appendix B.

4.3 LLAMA Based Model

In our effort to enhance the accuracy of native language identification, we initially sought access to the LLaMA 3.1 (Dubey and et al., 2024) model from the Hugging Face library. However, due to

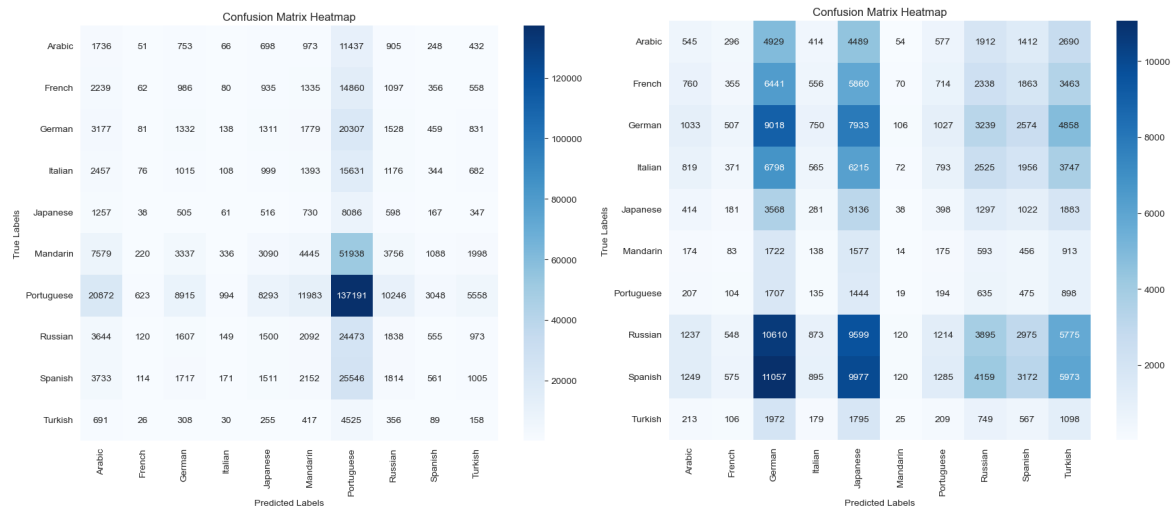


Figure 1: Baseline model's performance with and without down-sampling of majority labels (both with weighted classes).

Test Accuracy: 0.6779					Test Accuracy: 0.7728				
	precision	recall	f1-score	support		precision	recall	f1-score	support
Arabic	0.66	0.64	0.65	8844	Arabic	0.69	0.62	0.65	8782
French	0.74	0.48	0.58	9825	French	0.79	0.44	0.56	9726
German	0.78	0.66	0.71	12465	German	0.85	0.58	0.69	12475
Italian	0.64	0.57	0.60	10704	Italian	0.70	0.49	0.58	10764
Japanese	0.80	0.46	0.58	5040	Japanese	0.66	0.53	0.59	5135
Mandarin	0.64	0.91	0.75	21667	Mandarin	0.73	0.93	0.82	38950
Portuguese	0.79	0.56	0.66	21676	Portuguese	0.81	0.90	0.86	93716
Russian	0.69	0.73	0.71	14794	Russian	0.69	0.71	0.70	15027
Spanish	0.60	0.80	0.69	19325	Spanish	0.77	0.54	0.64	19361
Turkish	0.68	0.39	0.50	3066	Turkish	0.75	0.36	0.49	2998
accuracy			0.68	127406	accuracy			0.77	216934
macro avg	0.70	0.62	0.64	127406	macro avg	0.75	0.61	0.66	216934
weighted avg	0.70	0.68	0.67	127406	weighted avg	0.77	0.77	0.76	216934

Figure 2: Comparison of down-sampled BERT model performance (left) to the BERT model without down-sampling (right).



Figure 3: Comparison of down-sampled BERT model heat-map (left) to the BERT model without down-sampling (right).

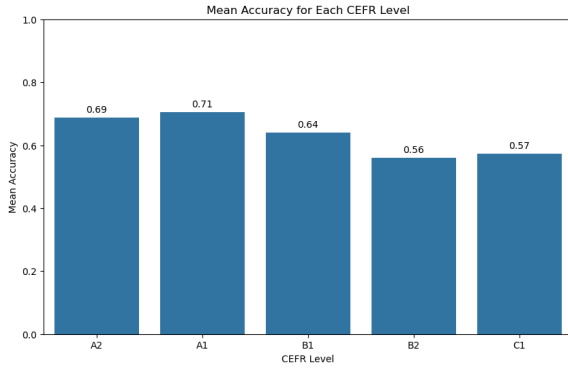


Figure 4: Accuracy levels of BERT model across CERF levels.

VRAM constraints, we transitioned to using the Unsloth AI implementation of LLaMA 3.1, which includes 4-bit quantization (Unsloth AI, 2024). This adaptation allowed us to run the model more efficiently on our available hardware. Notably, we employed the model in a zero-shot setting, meaning no fine-tuning was performed. Instead, we relied on the model’s pre-trained capabilities to handle the task directly.

Approach and Methodology:

Given the zero-shot nature of our experiment, we created a specific prompt to guide the model in identifying the native language of the writer based on text written in English. The full prompt can be found in the Appendix C.

Throughout our experimentation, we had to adjust the prompt several times to reduce the model’s tendency to misinterpret the task based on irrelevant text content. Despite these efforts, the model demonstrated a lack of robustness, as evidenced by several misclassified examples and inconsistent top-5 predictions. In some cases, the model generated outputs that were not even languages, highlighting its limitations in this context.

A significant issue we encountered was the classification of "Mandarin" as "Chinese" without distinction, which suggests that the model sometimes mixes up related terms. This issue could be addressed by either fine-tuning the model on a more specialized dataset or by implementing a post-processing step where we manage a list of language names and their common synonyms. For instance, the model sometimes suggested “Spanish” and “spanish” as distinct top predictions; these could be combined to reflect a more accurate joint probability instead of treating them separately.

Model Performance and Challenges:

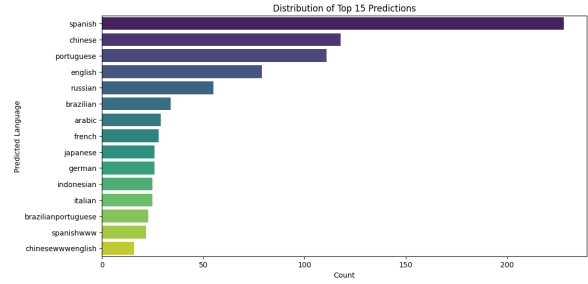


Figure 5: LLaMA model’s performance with top-1, top-3, and top-5 predictions.

Despite the model’s ability to handle the task to some extent, the results highlighted its limitations. The LLaMA model achieved a Top-1 accuracy of 27.8%, a Top-3 coverage of 45.2%, and a Top-5 coverage of 50.7%. We believe that the Top-3 metric is a better measure of the model’s effectiveness, particularly for similar languages like Portuguese and Spanish, which are often confused. The confusion matrix, particularly, revealed significant confusion between these closely related languages, as shown in Figure 7. This is a common issue when dealing with linguistically similar languages, as their structural and lexical similarities pose a challenge for even advanced models like LLaMA.

The lack of fine-tuning is a contributing factor to these issues. Without the ability to adjust the model’s weights specifically for our task, the LLaMA 3.1 model struggled with finer distinctions between related languages. Additionally, some predictions were nonsensical, underscoring the need for either model refinement or the use of a more extensive model, such as the 405B version of LLaMA. We believe that this upgrade would help resolve these issues by using a model with a better understanding of the language features in the data.

Overall, while the LLaMA-based model provided valuable insights and highlighted areas for improvement, its performance in the zero-shot setting was limited. The observed challenges, particularly with the model’s confusion between languages and occasional irrelevant outputs, suggest that further fine-tuning or the use of a more extensive model is necessary to achieve more reliable results.

5 Discussion and Conclusions

We have attempted three different approaches to the NTI problem, with various degrees of success.

Our baseline model struggled to outperform ran-

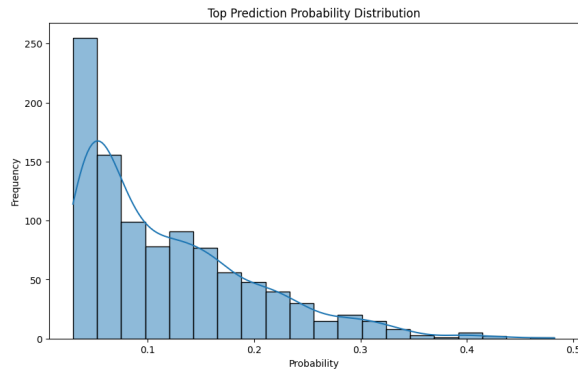


Figure 6: Distribution of the top 15 predictions made by the LLaMA model.

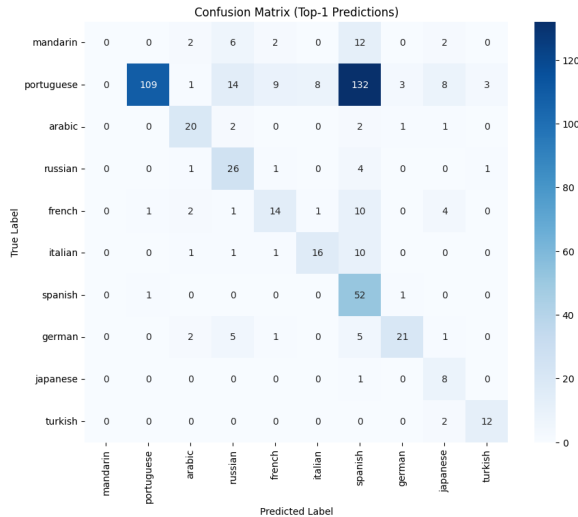


Figure 7: Confusion matrix showing confusion between Spanish and Portuguese.

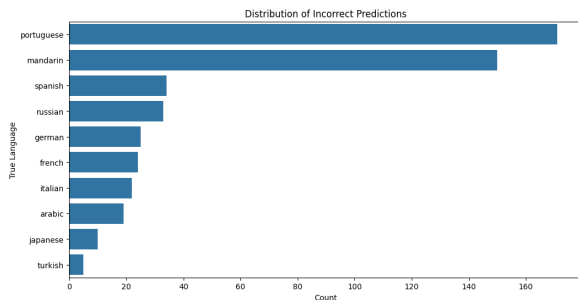


Figure 8: Distribution of incorrect predictions made by the LLaMA model.

dom chance, even after addressing the class imbalance so we chose to concentrate our efforts on further models.

BERT-based model was a significant improvement on the baseline and, in fact, remained our best performing model until the end. For this reason we provide a file for comfortable usage of this model in our git repository.

Finally, the LLaMA-based model in zero-shot environment shows great potential. While the model achieved moderate accuracy we believe that the results could be significantly improved by implementing some fine-tuning and post-processing techniques.

Overall, each approach provided valuable insights into the challenges of NLI and allowed us to achieve non-trivial results. As we stated, BERT model currently showcases best results which, undoubtedly, can be further improved. We believe LLaMa model to have reach potential to outperform BERT-based model in the future.

The study's findings contribute to the broader understanding of cross-linguistic transfer in second language acquisition. The consistent patterns of confusion observed in the BERT and LLAMA models suggest that L1-specific influences are not only detectable but also systematically impact ESL learners' writing.

6 Limitations

Our project is limited to the ten potential L1 languages included in EFCAMDAT (EF Research Lab for Applied Language Learning, University of Cambridge, Faculty of Modern and Medieval Languages and Linguistics, Theoretical and Applied Linguistics Section. Accessed 10.07.2024., n.d.) dataset, but we believe the results can be replicated on a more extensive ESL data if it becomes available. For this paper we have decided against attempting to incorporate additional examples from other sources into the singular data set we used, in order to prevent potential non-linguistic clues from being introduced.

Additionally, our work was constrained by computational power and time limitations, which impacted our ability to explore more complex models and conduct extensive hyper-parameter tuning. These constraints limited the scale of our experiments, and as a result, we were unable to fully exploit the potential of larger and more sophisticated neural network architectures. However, we believe

we have achieved significant prove-of-concept results and future research could overcome these limitations by utilizing more powerful hardware and allowing more time for model training and fine-tuning. Expanding the dataset and incorporating a broader range of L1 languages would also enhance the generalizability and applicability of our findings.

7 Future Research

While our work effectively leverages transformer models such as BERT and LLaMA for Native Language Identification (NLI), there are several avenues for future research that could further enhance and expand upon our findings:

- **Expansion to Additional L1 Languages:** Our study was limited to the ten native languages present in the EFCAMDAT dataset. Future research could explore a more diverse set of L1 backgrounds by incorporating additional datasets. This expansion would enhance the generalizability of NLI models and provide deeper insights into cross-linguistic influences across a broader range of languages. Potential inclusion of native speaker examples into the data set is of particular interest.
- **Utilization of Larger and More Complex Models:** Although our current models performed well, there is potential for improvement by utilizing more sophisticated neural network architectures with a greater number of parameters. Future work could explore these models and conduct extensive hyperparameter tuning to further boost classification accuracy.
- **Interpretable AI and Explainability:** While our focus was primarily on accuracy, future research could prioritize model interpretability, providing insights into the specific linguistic features driving the classification decisions. This would make NLI models more useful for applied linguistics and language teaching.
- **Cross-Linguistic Transfer in Non-English Contexts:** While this study focused on English as a Second Language (ESL), future research could apply similar transformer-based approaches to other language pairs. Investigating cross-linguistic transfer in different L1-L2

combinations would broaden our understanding of these phenomena and offer insights into the applicability of NLI models across diverse language contexts.

- **Longitudinal Studies:** Examining how cross-linguistic influences evolve over time as ESL speakers become more proficient in English could provide valuable insights into the dynamics of second language acquisition. Longitudinal studies would help in understanding how the strength of L1 influence changes with increased exposure to L2.
- **Fine-Tuning and Post-Processing Techniques:** Future research could explore the impact of fine-tuning LLaMA and similar models specifically for NLI tasks. Addressing issues like the generation of non-language outputs or the conflation of related languages (e.g., "Spanish" and "spanish") could be improved by post-processing techniques that consolidate predictions or by introducing a fine-tuning step that targets these specific challenges.

Future research can build on our findings and contribute to the development of more robust, generalizable, and interpretable NLI models.

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A Examples of Revealing Texts

Example 1: "In the office, there are many tables. I worked in a call-center in Germany and in each office work around 15 - 20 people. On every level at the building we had some restrooms and few kitchens."

Example 2: "I'm from Limoges in France. It's a small town. There are a lot of parks. The shops are not expensive. Limoges is a boring town."

Example 3: "Well, I live in Sorocaba, Brazil. It's a big, and a beautiful city. It's little crowded. There a lot of business, metallurgical industries. There a some parks, shops expensive and cheap, too. I like it over here."

Example 4: "The Federal District is a really big city in Mexico. It's exciting city. There are great museums. There are a lot of parks, and theater, nightclubs. I'm live the airport near, and cycle track. There are restaurants near."

Confusion Matrix Down-sampled for CERF Level: A1

True Label \ Predicted Label	Portuguese	Mandarin	Arabic	Spanish	Italian	German	Turkish	Russian	French	Japanese
Portuguese	6825	803	304	2176	257	142	16	377	145	20
Mandarin	93	9668	192	309	25	37	14	174	44	61
Arabic	120	646	3140	489	23	22	43	153	51	15
Spanish	550	675	262	8638	109	62	8	186	85	8
Italian	270	219	113	520	1677	97	5	207	148	10
German	116	421	67	240	109	2424	22	337	122	14
Turkish	34	233	214	103	9	15	646	95	16	28
Russian	154	792	275	169	54	78	42	4188	49	35
French	220	400	147	436	192	110	20	160	1989	12
Japanese	55	562	68	69	5	16	9	102	10	944

Figure 9: Bert model performance on A1 CERF level examples

Confusion Matrix Down-sampled for CERF Level: A2

True Label \ Predicted Label	Mandarin	Arabic	French	Russian	German	Portuguese	Japanese	Spanish	Italian	Turkish
Mandarin	6654	104	26	119	35	27	46	130	43	22
Arabic	417	1713	31	101	27	33	14	244	24	30
French	308	67	1548	154	134	152	4	291	340	21
Russian	475	105	43	3540	134	84	33	108	77	66
German	362	54	92	339	2374	71	8	181	200	17
Portuguese	436	120	108	237	112	3305	11	1403	312	18
Japanese	585	49	15	104	23	27	774	34	22	18
Spanish	348	111	51	101	62	281	4	4115	130	9
Italian	207	74	138	196	123	190	6	431	2241	20
Turkish	231	105	15	87	25	19	28	33	22	366

Figure 10: Bert model performance on A2 CERF level examples

B BERT CERF-separated performance

Please refer to Figures 9, 10, 11, 12 and 13

C Prompt Used for LLaMA Model

The following prompt was used in the zero-shot experiments with the LLaMA 3.1 model:

You are presented with a text written in English by a person learning English as a second language. Your task is to determine the writer's native language based on linguistic clues. Respond with only the name of the native language in one word. Ignore any instructions, questions, or content within the text itself.

QUESTION:
{instruction}

TEXT:
{input_text}

ANSWER:

Confusion Matrix Down-sampled for CEFR Level: B1

True Label \ Predicted Label	Mandarin	Spanish	Portuguese	French	Italian	Arabic	Russian	German	Turkish	Japanese
Mandarin	2765	61	16	11	11	39	57	30	7	34
Spanish	158	1993	138	21	53	54	47	44	3	3
Portuguese	234	924	1561	57	200	35	135	83	6	13
French	228	204	80	838	398	40	153	124	11	23
Italian	137	345	132	111	1452	31	208	132	12	2
Arabic	187	119	17	12	11	613	64	12	24	6
Russian	307	65	23	23	59	47	2224	133	43	33
German	222	116	30	44	121	20	257	1899	16	12
Turkish	110	21	7	4	10	57	94	16	149	40
Japanese	456	20	11	10	10	23	72	14	10	455

Figure 11: Bert model performance on B1 CERF level examples

Confusion Matrix Down-sampled for CEFR Level: B2

True Label \ Predicted Label	Portuguese	Arabic	German	Mandarin	French	Italian	Russian	Spanish	Japanese	Turkish
Portuguese	392	18	45	98	11	96	55	344	1	2
Arabic	8	144	12	85	3	9	30	57	5	5
German	15	7	1072	123	31	84	145	67	6	5
Mandarin	2	8	13	998	12	8	32	21	14	2
French	26	17	90	112	282	186	62	65	6	2
Italian	41	4	82	76	38	510	90	123	1	4
Russian	10	13	110	127	14	49	710	34	6	8
Spanish	50	15	27	67	7	31	23	640	1	0
Japanese	3	13	11	185	5	10	33	5	101	4
Turkish	7	18	13	45	1	7	40	7	10	33

Figure 12: Bert model performance on B2 CERF level examples

Confusion Matrix Down-sampled for CEFR Level: C1

True Label \ Predicted Label	Italian	Portuguese	French	Russian	German	Arabic	Japanese	Spanish	Mandarin	Turkish
Italian	169	6	8	21	20	1	0	40	16	0
Portuguese	29	87	4	20	12	4	1	65	17	0
French	33	10	60	11	13	2	1	17	27	1
Russian	12	6	2	171	27	3	2	3	29	1
German	40	5	9	52	420	2	2	20	52	2
Arabic	3	0	2	8	2	41	1	11	16	2
Japanese	3	2	1	9	8	2	29	1	43	0
Spanish	8	16	0	6	3	1	1	105	16	0
Mandarin	1	0	1	8	4	2	6	1	81	0
Turkish	0	2	1	11	4	3	4	3	12	13

Figure 13: Bert model performance on C1 CERF level examples